

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	6	9	20

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

$$\text{let } Y = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{pmatrix}, \quad \Theta = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}, \quad \epsilon = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix}$$

From above equation

$$Y = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} = X\Theta + \epsilon$$

We know that

$$\hat{\Theta} = (X^T X)^{-1} \cdot X^T \cdot Y \rightarrow \text{least square Estimate.}$$

$$(X^T X) = \begin{pmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1+1+4 & 1-2-2 \\ 1-2-2 & 1+4+1 \end{pmatrix} = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}$$

$$(X^T X)^{-1} = \frac{\text{Adj}(X^T X)}{|X^T X|}$$

$$\text{Adj}(X^T X) = \begin{pmatrix} c_{22} & -c_{21} \\ -c_{12} & c_{11} \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 3 & 6 \end{pmatrix}$$

$$|X^T X| = \begin{vmatrix} 6 & -3 \\ -3 & 6 \end{vmatrix} = 36 - 9 = 27$$

$$\Rightarrow (X^T X)^{-1} = \frac{1}{27} \begin{pmatrix} 6 & 3 \\ 3 & 6 \end{pmatrix}$$

$$(X^T X)^{-1} \cdot X^T = \frac{1}{27} \begin{pmatrix} 6 & 3 \\ 3 & 6 \end{pmatrix}_{2 \times 2} \begin{pmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{pmatrix}_{2 \times 3}$$

$$= \frac{1}{27} \begin{pmatrix} 6+3 & 6-6 & 12-3 \\ 3+6 & 3-12 & 6-6 \end{pmatrix}$$

$$= \frac{1}{27} \begin{pmatrix} 9 & 0 & 9 \\ 9 & -9 & 0 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

$$(X^T X)^{-1} \cdot X^T \cdot Y = \frac{1}{3} \begin{pmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} y_1 + y_3 \\ y_1 - y_2 \end{pmatrix}$$

$$\Rightarrow \hat{\theta} = \begin{pmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} y_1 + y_3 \\ y_1 - y_2 \end{pmatrix}$$

$$\hat{\theta}_1 = \frac{y_1 + y_3}{3}$$

$$\hat{\theta}_2 = \frac{y_1 - y_2}{3}$$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
- (a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.
- (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

$$X_1 = U_1, \quad X_2 = U_2, \quad X_3 = U_1 + U_2$$

$$U_1, U_2 \sim U(0, 1)$$

(a) Let
$$\rho = \begin{bmatrix} \rho_{11} & \rho_{12} & \rho_{13} \\ \rho_{21} & \rho_{22} & \rho_{23} \\ \rho_{31} & \rho_{32} & \rho_{33} \end{bmatrix}$$

Where, ρ_{ij} = Correlation between X_i, X_j

W.K.T
$$\rho_{ij} = \frac{\text{Cov}(X_i, X_j)}{\sqrt{\text{Cov}(X_i, X_i) * \text{Cov}(X_j, X_j)}}$$

Given that $U_1, U_2 \Rightarrow$ independent random variables

$$\Rightarrow \text{Cov}(U_1, U_2) = 0 \rightarrow \textcircled{1}$$

$$\rho_{ii} = \frac{\text{Cov}(X_i, X_i)}{\sqrt{\text{Cov}(X_i, X_i) * \text{Cov}(X_i, X_i)}} = 1 \quad \forall \quad i = 1, 2, 3$$

$$\rho_{ij} = \rho_{ji} \quad \text{from symmetry}$$

$$\rho_{12} = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Cov}(X_1, X_1)} \cdot \sqrt{\text{Cov}(X_2, X_2)}} \xrightarrow{\text{Cov}(U_1, U_2) = 0 \text{ from eqn } \textcircled{1}}$$

$$= 0$$

$$\rho_{23} = \frac{\text{Cov}(U_1, U_1 + U_2)}{\sqrt{\text{Cov}(U_1, U_1)} \cdot \sqrt{\text{Cov}(U_1 + U_2, U_1 + U_2)}}$$

$$\text{Cov}(U_1, U_1) = \text{Cov}(U_2, U_2) = \frac{1}{12} \quad \left[\because \text{Var}(U_1) = \text{Var}(U_2) = \frac{(b-a)^2}{12}, \right.$$

$$b=1, a=0$$

for uniform R.V

$$\rho_{13} = \frac{\text{Cov}(U_1, U_1 + U_2)}{\sqrt{\text{Cov}(U_1, U_1)} \cdot \sqrt{\text{Cov}(U_1 + U_2, U_1 + U_2)}}$$

$$\begin{aligned} \text{Cov}(U_1 + U_2, U_1 + U_2) &= \text{Cov}(U_1, U_1) + \text{Cov}(U_1, U_2) + \text{Cov}(U_2, U_1) + \text{Cov}(U_2, U_2) \\ &= \left(\frac{1}{12}\right) + 0 + 0 + \left(\frac{1}{12}\right) \\ &= \left(\frac{1}{6}\right) \end{aligned}$$

$$\rho_{13} = \frac{\left(\frac{1}{12}\right)}{\sqrt{\frac{1}{12} \cdot \frac{1}{6}}} = \left(\frac{1}{\sqrt{2}}\right)$$

similarly

$$\rho_{23} = \frac{1}{\sqrt{2}}$$

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hence

$$\rho = \begin{pmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{pmatrix}$$

correlation matrix of x .

(b)

let us find $(\lambda_1, \lambda_2, \lambda_3)$ eigen values of matrix ρ .

$$|P - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1-\lambda & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1-\lambda \end{vmatrix} = 0$$

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$$\Rightarrow (1-\lambda) \left((1-\lambda)^2 - \frac{1}{2} \right) + \frac{1}{\sqrt{2}} \left(0 - \frac{(1-\lambda)}{\sqrt{2}} \right) = 0$$

$$\Rightarrow (1-\lambda)^2 - (1-\lambda) = 0$$

$$\Rightarrow (1-\lambda) [(1-\lambda) - 1] = 0$$

$$\Rightarrow (1-\lambda)(-\lambda)(2-\lambda) = 0$$

$$\Rightarrow \lambda_1 = 2, \lambda_2 = 0, \lambda_3 = 0$$

Variance of first PC
is $\lambda_{\max}(\lambda_1, \lambda_2, \lambda_3)$

$$= 2$$

3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

- (a) (6 points) Cluster the four items using each of the following procedures:
- Complete linkage hierarchical procedure.
 - Average linkage hierarchical procedure.
- (b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

Solⁿ.

(a) (i) Complete linkage

$$d(u,v,w) = \max \{d_{u,w}, d_{v,w}\}$$

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

Combine 1,3

	(13)	2	4
(13)	0		
2	2	0	
4	5	3	0

$$d_{(13)2} = \max \{d_{12}, d_{32}\} = \max \{1, 2\} = 2$$

$$d_{(13)4} = \max \{d_{14}, d_{34}\} = \max \{5, 3.5\} = 5$$

(134)	2
0	
2	0
5	

$$d_{(134)2} = \max \{d_{(13)2}, d_{(13)4}\} = \max \{2, 5\} = 5$$

$$1342 \begin{bmatrix} 0 \end{bmatrix}$$

(ii) Average linkage

$$d(u,v), w = \frac{\sum_{i=1}^{N(u)} d_{ij} + \sum_{j=1}^{N(w)} d_{ij}}{N(u) + N(w)}$$

$N(A)$ = Number of variables in the grouping A.

given matrix

	1	2	3	4
1	0			
2	1	0		
3	2	0	0	
4	5	3	3.5	0

$d_{1,2} =$

↓

	(1,2)	3	4
(1,2)	0		
3		0	
4		3	0

$$d_{(1,2),3} = \frac{d_{1,2} + d_{3,2}}{(2) + 1}$$

$$= \frac{1 + 2}{2} = 1.5$$

$$d_{(1,2),4} = \frac{d_{1,4} + d_{3,4}}{2 + 1}$$

$$= \frac{(5 + 3.5)}{2} = \frac{8.5}{2} = 4.25$$

Done in Next Page,,

Solution

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	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (i) Complete linkage

$$d(u,v,w) = \max \{ d_{u,w} ; d_{v,w} \}$$

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0



	(12)	3	4
(12)	0		
3	11	0	
4	5	3.5	0

$$d_{(12),3} = \max \{ d_{1,3}, d_{2,3} \}$$

$$= \max \{ 11, 2 \} = 11$$

$$d_{(12),4} = \max \{ d_{1,4}, d_{2,4} \}$$

$$= \max \{ 5, 3 \} = 5$$

	(12)	(34)
(12)	0	
(34)	11	0



(1234)

(1234)	0
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$$d_{(12),(34)} = \max \{ d_{(12),3}, d_{(12),4} \}$$

$$= \max \{ 11, 5 \} = 11$$

(b) Average linkage

$$d(w,w) = \frac{\sum_{i=1}^{N(u)} \sum_{j=1}^{N(w)} d_{ij}}{N(u) \cdot N(w)}$$

$N(x) \Rightarrow$ Number of variables in set " x ".

$$\begin{array}{c}
 1 \quad 2 \quad 3 \quad 4 \\
 \begin{bmatrix}
 0 & & & \\
 1 & 0 & & \\
 11 & 2 & 0 & \\
 5 & 3 & 3.5 & 0
 \end{bmatrix}
 \end{array}
 \rightarrow
 \begin{array}{c}
 (12) \quad 3 \quad 4 \\
 \begin{bmatrix}
 0 & & \\
 6.5 & 0 & \\
 4 & 3.5 & 0
 \end{bmatrix}
 \end{array}$$

$$d_{(12),3} = \frac{d_{1,3} + d_{2,3}}{2} = \frac{11+2}{2} = 6.5$$

$$d_{(12),4} = \frac{d_{1,4} + d_{2,4}}{2} = \frac{5+3}{2} = 4$$

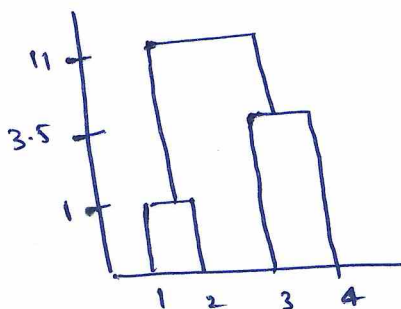
$$\begin{array}{c}
 (12) \quad (34) \\
 \begin{bmatrix}
 0 & \\
 5.25 & 0
 \end{bmatrix}
 \end{array}
 \rightarrow
 \begin{array}{c}
 (1234) \\
 [0]
 \end{array}$$

$$\begin{aligned}
 d_{(12),(34)} &= \frac{d_{(12),3} + d_{(12),4}}{2} \\
 &= \left(\frac{6.5+4}{2} \right) = \frac{10.5}{2} \\
 &= 5.25
 \end{aligned}$$

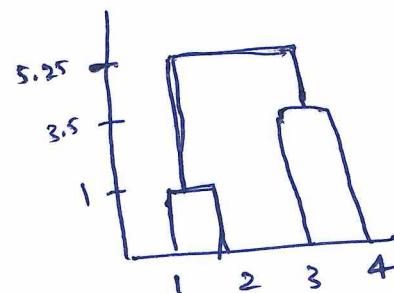
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(b) Dendrograms.

(i) Complete linkage



(ii) Average linkage



* Initially both type of methods with same separation.

cluster (1,2) (3,4) & later

* By average linkage distance between group (12), (3,4) is less than complete linkage procedure.

1. The following are the main features of the Indian legal system:

- (i) The Indian legal system is a unitary system.
- (ii) The Indian legal system is a common law system.
- (iii) The Indian legal system is a civil law system.
- (iv) The Indian legal system is a mixed system.

2. The following are the main sources of law in India:

- (i) The Constitution of India.
- (ii) The Acts of Parliament.
- (iii) The State Laws.
- (iv) The Custom and Usage.
- (v) The Judicial Decisions.

3. The following are the main branches of law in India:

- (i) Criminal Law.
- (ii) Civil Law.
- (iii) Administrative Law.
- (iv) Constitutional Law.
- (v) International Law.

4. The following are the main principles of law in India:

- (i) The principle of equality.
- (ii) The principle of justice.
- (iii) The principle of fairness.
- (iv) The principle of proportionality.
- (v) The principle of reasonableness.

5. The following are the main functions of the law in India:

- (i) To maintain order and discipline.
- (ii) To protect the rights and liberties of the citizens.
- (iii) To promote the welfare of the community.
- (iv) To resolve disputes and conflicts.
- (v) To enforce the law and maintain the rule of law.